

uses volatility or variance units. The mean-variance frontier for the United States and Japan represents all combinations of the United States and Japan. Naturally, the square representing the United States is a 100% U.S. portfolio, and the circle representing Japan is a 100% Japanese portfolio. All the other positions on the mean-variance frontier represent portfolios containing different amounts of the United States and Japan.

The mean-variance frontier is a *parabola*, or a bullet. The top half of the mean-variance frontier is *efficient*: an investor cannot obtain a higher reward, or expected return, for a given level of risk measured by volatility. Investors will choose portfolios on the top, efficient part of the frontier. The United States sits on the underbelly of the bullet. You can achieve a higher expected return for the same volatility by moving onto the top half of the frontier. The United States is an *inefficient* portfolio. *No one should hold a 100% U.S. portfolio.*

2.2. DIVERSIFICATION

In the *Merchant of Venice*, Shakespeare tells us why we should diversify:

... I thank my fortune for it,
My ventures are not in one bottom trusted,
Nor to one place; nor is my whole estate
Upon the fortune of this present year:
Therefore my merchandise makes me not sad.

This is spoken by the merchant Antonio, who diversifies so that all of his risk is not tied up in just one ship (“one bottom”).⁶ Similarly, we don’t want to bet everything on just the United States or Japan in isolation (“one place”).

The fact that Japanese equities have moved differently from U.S. equities, especially over the 1980s and 1990s, causes the mean-variance frontier to bulge outwards to the left. The correlation between the United States and Japan is 35% over the sample. (The United States–Japan correlation post-2000 is 59%, still far below one.) Mean-variance frontiers are like the Happy (or Laughing) Buddha: the fatter the stomach or bullet, the more prosperous the investor becomes. Notice that the left-most point on the mean-variance frontier in Figure 3.2 has a lower volatility than either the United States or Japan. This portfolio, on the left-most tip of the bullet, is called the *minimum variance portfolio*.

⁶ Although Antonio owns many ships, his wealth is held in only one asset class—venture capital—and so is undiversified across asset classes and is illiquid. Antonio should diversify across factors (see chapter 14). Unable to raise cash to lend to his friend, he is forced to go to Shylock for a loan, which he is forced to pay with a “pound of flesh” when he defaults. I cover asset allocation with illiquid assets in chapter 13.